GOLDEN SNITCH

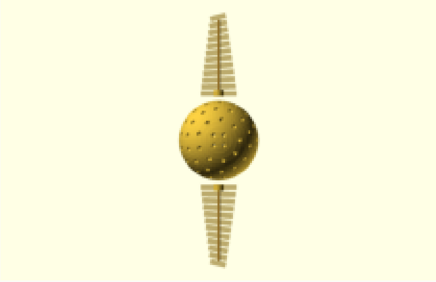
Walnut-sized evasion drone — beamed-power, full-match flight

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

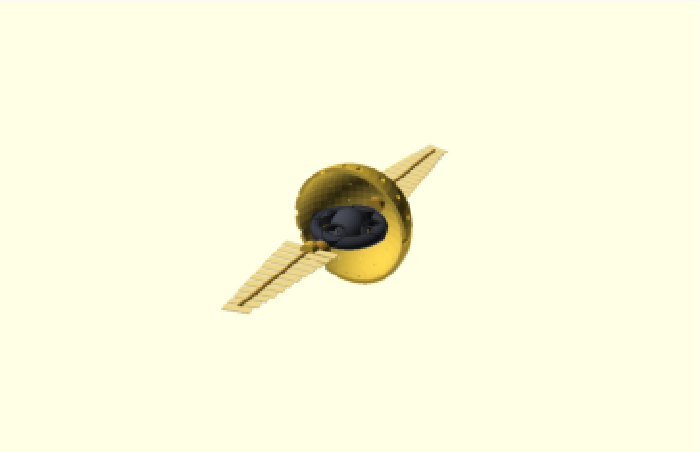
ISOMETRIC

SIDE

TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius `ball/params.py snitch.radius = 0.04 m`	**40 mm (Ø80)**
Mass target `snitch.mass = 0.05 kg`	**50 g (kept)**
Top speed `snitch.max_speed`	**19 m/s**
Max accel `snitch.max_accel (absurdly agile)`	**32 m/s²**
Lift `swarm core — the real flight`	**4 × micro-EDF (internal)**
Battery sprint `1S 300 mAh alone — ≈ the fatigue_tau`	**~2.7 min**
Match endurance `analysis/snitch_endurance.py`	**continuous (beamed)**
Pitch power `5.8 GHz steered to the broadcast pose`	**~2.5 kW array**

NOTE

WINGS ARE THEATRE (SNT-02/03/04 flutter; lift is the internal EDF swarm SNT-05 — don't tell Potterheads). A 50 g walnut can't carry fuel like the broom, so it HARVESTS energy instead: the gold dimple-vents double as a 5.8 GHz rectenna (SNT-15/16) and a pitch array beams power onto its broadcast pose (SNT-19/20). The cell becomes a buffer (jukes on a supercap SNT-17). Form + agility kept, match filled, no pit.

BILL OF MATERIALS — SNT (20 line items, prototype qty 1)

REF	PART	SPEC	QTY	$ EA	$ EXT
SNT-01	Gold-clad vented sphere shell	Ø80 mm (R=40 mm), walnut-sized, dimple vents	1	90	90
SNT-02	Animatronic flapping wing ass...	thin filigree vane on hinge, span ~76 mm	2	28	56
SNT-03	Coreless gearmotor + crank	6 mm coreless, flap linkage, 8-14 Hz beat	2	9	18
SNT-04	Wing membrane	5 micron gold-tint mylar, feather-cut	2	4	8
SNT-05	Micro ducted-fan lift unit	Ø16 mm internal, the ACTUAL lift	4	22	88
SNT-06	Micro brushless motor	4 mm, paired to micro-EDF	4	11	44
SNT-07	Micro ESC (4-in-1)	5 A, integrated	1	26	26
SNT-08	Nano flight controller	STM32 class, IMU on-board, predictive evasion	1	38	38
SNT-09	Optical-flow + ToF sensor	360deg-ish reach detection	2	16	32
SNT-10	Micro LiDAR / proximity ring	detects incoming hands/airframes	1	45	45
SNT-11	Capacitive capture sensor	registers an enclosing hand	1	12	12
SNT-12	Micro Li-Po cell	1S 300 mAh, now a ride-through BUFFER (not the engine)	1	8	8
SNT-13	Gold micro-LED	glow so the crowd can track it	1	5	5
SNT-14	2.4 GHz telemetry link	reports pose to the Dementor referee	1	14	14
SNT-15	Conformal rectenna array	5.8 GHz, printed into the gold dimple-vent pattern	1	34	34
SNT-16	RF->DC harvester PMIC + MPPT	rectifier + maximum-power tracking, 5.8 GHz	1	17	17
SNT-17	Graphene supercapacitor	7 F @ 5.4 V, ~1-2 g — the PEAK buffer	1	12	12
SNT-18	In-shell power-bus + balancing	rectenna / supercap / cell tri-source manager	1	9	9
SNT-19	Beam-steer retro-beacon	lets the pitch array lock its spot onto the Snitch	1	11	11
SNT-20	Pitch power-beaming array (pe...	5.8 GHz phased array, ~6 m^2 aperture, ~2.5 kW wall	1	0	0

UNIT TOTAL (prototype): $567

DRAWN	DATE	UNITS	SCALE	PART	REV	CAD
NIMBUS HW	2026-06-27	mm / SI	NTS	SNT	v0.1	cad/snitch.scad